STA2002 Tutorial 7

1. Let $W_1 < W_2 < \cdots < W_n$ be the order statistics of n independent observations from a $U(0,1)$ distribution.
   (a) Find the pdf of W_1, W_n and $W_r, r = 1, \ldots n$.
   (b) Show that W_r is a beta variable.
   (c) Show that $E(W_r) = r/(n+1), E(W_r^2) = r(r+1)/(n+1)(n+2)$

   Solution:

   (a)

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!1!(n-k)!}[F(x)]^{k-1} f(x)[1-F(x)]^{n-k}$$
$$g_1(w) = n[1-w]^{n-1}, \quad 0 < w < 1$$
$$g_n(w) = n[w]^{n-1}, \quad 0 < w < 1$$
$$g_r(w) = \frac{n!}{(r-1)!(n-r)!}[w]^{r-1}[1-w]^{n-r}, \quad 0 < w < 1$$

   (b) Let $w = w_r$. The pdf of W_r is

$$\begin{aligned} g_r(w) &= \frac{n!}{(r-1)!(n-r)!}[w]^{r-1}[1-w]^{n-r} \cdot 1 \\ &= \frac{\Gamma(r+n-r+1)}{\Gamma(r)\Gamma(n-r+1)} w^{r-1}(1-w)^{n-r+1-1} \end{aligned}$$

   The pdf of beta distribution: $f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1$ Thus W_r has a beta distribution with $\alpha = r, \beta = n-r+1$ (c)

$$\begin{aligned} E(W_r) &= \int_0^1 w \frac{n!}{(r-1)!(n-r)!} w^{r-1}(1-w)^{n-r} dw \\ &= \left(\frac{r}{n+1}\right) \int_0^1 \frac{(n+1)!}{r!(n-r)!} w^r (1-w)^{n-r} dw \\ &= \frac{r}{n+1} \\ E(W_r^2) &= \int_0^1 w^2 \frac{n!}{(r-1)!(n-r)!} w^{r-1}(1-w)^{n-r} dw \\ &= \frac{r(r+1)}{(n+2)(n+1)} \int_0^1 \frac{(n+2)!}{(r+1)!(n-r)!} w^{r+1}(1-w)^{n-r} dw \\ &= \frac{r(r+1)}{(n+2)(n+1)} \end{aligned}$$

2. Let $Y_1 < Y_2 < \cdots < Y_8$ be the order statistics of eight independent observations $X_1, X_2, \cdots, X_8$ from a continuous-type distribution with 70 th percentile $\pi_{0.7} = 27.3$
(a) Determine $P(Y_7 < 27.3)$.
(b) Find $P(Y_5 < 27.3 < Y_8)$.

Solution:

(a) Let W denote the number of X_i which is less or equal to 27.3.
Since $P(X_i < 27.3) = 0.7$, therefore,

$$\begin{aligned} P(Y_7 < 27.3) &= P(W \geq 7) = P(W = 7) + P(W = 8) \\ &= \binom{8}{7} (0.7)^7 (0.3) + \binom{8}{8} (0.7)^8 = 0.2553 \end{aligned}$$

(b)

$$\begin{aligned} P(Y_5 < 27.3 < Y_8) &= P(5 \leq W \leq 7) \\ &= P(W = 5) + P(W = 6) + P(W = 7) \\ &= \binom{8}{5} (0.7)^5 (0.3)^3 + \binom{8}{6} (0.7)^6 (0.3)^2 + \binom{8}{7} (0.7)^7 (0.3) = 0.7483 \end{aligned}$$

3. Let m denote the median weight of " 80 -pound" bags of water softener pellets. Use the following random sample of n = 14 weights to find an approximate 95% confidence interval for m :

80.51	80.28	80.40	80.35	80.38	80.28	80.27
80.16	80.59	80.56	80.32	80.27	80.53	80.32

(a) Find a 94.26% confidence interval for m.
(b) The interval (y_6, y_{12}) could serve as a confidence interval for $\pi_{0.6}$. What is its confidence coefficient?

Solution: (a) $(y_4 = 80.28, y_{11} = 80.51)$ is a 94.26% confidence interval for m. (b) $(y_6 = 80.32, y_{12} = 80.53)$;

$$\sum_{k=6}^{11} \binom{14}{k} (0.6)^k (0.4)^{14-k} = \sum_{3}^{8} \binom{14}{k} (0.4)^k (0.6)^{14-k} = 0.9417 - 0.0398 = 0.9019$$

The interval is $(y_6 = 80.32, y_{12} = 80.53)$.

4. (**Supplementary**) Let m equal the median of the posttest grip strengths in the right arms of male freshmen in a study of health dynamics. We shall use observations on n = 15 such students to test the null hypothesis $H_0 : m = 50$ against the alternative hypothesis $H_1 : m > 50$. (a) Using the Wilcoxon statistic, define a critical region that has an approximate significance level of $\alpha = 0.05$.
(b) Given the observed values
58.0 52.5 46.0 57.5 52.0 45.5 65.5 71.0 57.0 54.0 48.0 58.0 35.5 44.0 53.0
what is your conclusion?
(c) What is the p-value of this test?

(a) The critical region is given by

$$w \geq 1.645\sqrt{15(16(31)/6)} = 57.9$$

(b) In the following display, those differences that were negative are underlined.

$\lvert x_i - 50\rvert$	2	$\underline{2}$	2.5	3	4	$\underline{4}$	$\underline{4.5}$	$\underline{6}$	7	7.5	8	8	$\underline{14.5}$	15.5	21
Ranks:	1.5	1.5	3	4	5.5	5.5	7	8	9	10	11.5	11.5	13	14	15

The value of the Wilcoxon statistics is

$$w = 1.5 - 1.5 + 3 + 4 + 5.5 - 5.5 - 7 - 8 + 9 + 10 + 11.5 + 11.5 - 13 + 14 + 15 = 50$$

Since

$$z = \frac{50}{\sqrt{15(16)(31)/6}} = 1.420 < 1.645,$$

or since $w = 50 < 57.9$, we don not reject H_0. (c) The approximate p-value is, using the one-unit correction.

$$P(W \geq 50) = P(W \geq 50) \approx P\left(Z \geq \frac{49}{\sqrt{15(16)(31)/6}}\right) = P(Z \geq 1.3915) = 0.082$$